Integration by parts (7.1)



The product rule for derivatives says

$$\frac{d}{dx}\left(f(x)g(x)\right) = f(x)g'(x) + f'(x)g(x)$$

$$\int \frac{d}{dx}\left(f(x)g(x)\right)dx = \int f(x)g'(x)\,dx + \int f'(x)g(x)\,dx$$

$$f(x)g(x) + C = \int f(x)g'(x)\,dx + \int f'(x)g(x)\,dx$$

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int f'(x)g(x)\,dx$$

"+C" hidden in here

Integration by parts: For any functions $f(x), g(x)$

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx$$

$$\int x\cos x\,dx = x\sin x - \int \sin x\,dx$$

$f(x) = x \quad g'(x) = \cos x$

$f'(x) = 1 \quad g(x) = \sin x$

$$= x\sin x - (-\cos x + C)$$

$$= x\sin x + \cos x + C$$

Formula holds for any g, so we can choose any antiderivative of g'

Integration by parts: For any functions $f(x)$, $g(x)$

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx$$

$u = f(x) \quad du = f'(x)\,dx$

$v = g(x) \quad dv = g'(x)\,dx$

$$\int u\,dv = uv - \int v\,du$$

Integration by parts: For functions $u = f(x)$ and $v = g(x)$,

$$\int u\,dv = uv - \int v\,du$$

$$\int x\cos x\,dx = x\sin x - \int \sin x\,dx$$

$u = x \quad dv = \cos x\,dx$

$du = dx \quad \int dv = \int \cos x\,dx$

$v = \sin x + C$

Can choose any antiderivative

$x\sin x - (-\cos x + C)$

$x\sin x + \cos x + C$

$$\int xe^{2x}\,dx = \frac{1}{2}xe^{2x} - \int \frac{1}{2}e^{2x}\,dx$$

$u = x \quad dv = e^{2x}\,dx$

$du = dx \quad v = \frac{1}{2}e^{2x}$

$$= \frac{1}{2}xe^{2x} - \left(\frac{1}{4}e^{2x} + C\right)$$

$$= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$$

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} dx$$

$u = \ln x \quad dv = dx$

$du = \frac{1}{x} dx \quad v = x$

$$= x \ln x - \int dx$$

$$= x \ln x - (x + C)$$

$$= x \ln x - x + C$$

Integration by parts: For functions $u = f(x)$ and $v = g(x)$, defined on $[a, b]$,

$$\int_a^b u\,dv = uv\Big|_a^b - \int_a^b v\,du$$

$$u(b)v(b) - u(a)v(a)$$

$$\int_1^e x \ln x \, dx = \frac{1}{2}x^2 \ln x \Big|_1^e - \int_1^e \frac{1}{2}x^2 \cdot \frac{1}{x} dx$$

$u = \ln x \quad dv = x\,dx$

$du = \frac{1}{x}dx \quad v = \frac{1}{2}x^2$

$$\frac{1}{2}x^2 \ln x \Big|_1^e = \frac{1}{2}e^2 \ln e - \frac{1}{2}\ln 1$$

$$= \frac{1}{2}e^2$$

$$\int_1^e \frac{1}{2}x^2 \cdot \frac{1}{x} dx = \frac{1}{2}\int_1^e x\,dx = \frac{1}{4}x^2 \Big|_1^e = \frac{1}{4}e^2 - \frac{1}{4}$$

$$\frac{1}{2}e^2 - \left(\frac{1}{4}e^2 - \frac{1}{4}\right) = \frac{1}{4}e^2 + \frac{1}{4}$$

$$\int_0^1 x^2 e^x \, dx$$

First evaluate

$$\int x^2 e^x \, dx = x^2 e^x - \underbrace{\int 2x e^x \, dx}_{2\int x e^x \, dx}$$

$u = x^2 \quad dv = e^x \, dx$

$du = 2x \, dx \quad v = e^x$

$$\int x e^x \, dx = xe^x - \int e^x \, dx = \boxed{xe^x - e^x + C}$$

$u = x \quad dv = e^x dx$

$du = dx \quad v = e^x$

$$\int x^2 e^x dx = x^2 e^x - 2\int x e^x dx$$

$$= x^2 e^x - 2(xe^x - e^x + C)$$

$$= x^2 e^x - 2xe^x + 2e^x + C$$

$$\int_0^1 x^2 e^x dx = x^2 e^x - 2xe^x + 2e^x \Big|_0^1$$

$$= (e - 2e + 2e) - (0 - 0 + 2)$$

$$= e - 2$$

$$\int e^x \sin x \, dx = -e^x \cos x - \int (-\cos x) e^x \, dx$$

$u = e^x \quad dv = \sin x \, dx$

$du = e^x \, dx \quad v = -\cos x$

$$= -e^x \cos x + \int e^x \cos x \, dx$$

$$\int e^x \cos x \, dx = e^x \sin x - \int e^x \sin x \, dx$$

$u = e^x \quad dv = \cos x \, dx$

$du = e^x \, dx \quad v = \sin x$

$$\int e^x \sin x \, dx = -e^x \cos x + \int e^x \cos x \, dx$$

$$\int e^x \cos x \, dx = e^x \sin x - \int e^x \sin x \, dx$$

$$\int e^x \sin x \, dx = -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx$$

$$2\int e^x \sin x \, dx = e^x (\sin x - \cos x)$$

$$\int e^x \sin x \, dx = \frac{1}{2} e^x (\sin x - \cos x) + C$$

$$\int_0^1 \tan^{-1} x \, dx$$

$u = \tan^{-1} x \quad dv = dx$

$du = \frac{1}{1+x^2} dx \quad v = x$

$$\int \tan^{-1} x \, dx = x \tan^{-1} x - \underbrace{\int \frac{x}{1+x^2} dx}$$

$$\int \frac{x}{1+x^2} dx = \int \frac{1}{2} \frac{1}{u} du = \frac{1}{2} \ln|u| + C$$

$u = 1+x^2 \quad du = 2x \, dx$

$$= \frac{1}{2} \ln(1+x^2) + C$$

$$\int \tan^{-1} x\,dx = x\tan^{-1}x - \frac{1}{2}\ln(1+x^2) + C$$

$$\int_0^1 \tan^{-1} x\,dx = \left(x\tan^{-1}x - \frac{1}{2}\ln(1+x^2)\right)\Big|_0^1$$

$$= \left(\frac{\pi}{4} - \frac{1}{2}\ln 2\right) - (0 - 0)$$

$$= \frac{\pi}{4} - \frac{1}{2}\ln 2$$